Rohan Shenoy

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Education

University of California, Berkeley Aug 2022 – May 2026
Bachelor of Arts in Mathematics and Computer Science (Double Major) *GPA: 3.8/4.0*
Relevant Coursework: Machine Learning, Probability Theory and Random Processes, Financial Economics, Optimization Models, Efficient Algorithms and Intractable Problems, Operating Systems, Computer Architecture, Data Structures, Real Analysis, Numerical Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, Computer Vision

Experience

Research Intern Apr 2024 – Present
Massachusetts Institute of Technology *Cambridge, MA*
- Developed a Gymnasium simulator environment for nuclear spin dynamics; accelerated RL training runtime by ∼1000x using parallelized computation, batched gradient updates, and optimized integration routines
- Pioneered an RL control approach for NMR-based atomic measurement; trained a PPO agent with tuned policy network, experimenting with reward shaping, and teacher forcing to stabilize learning, reducing scan time by 16%
- Architected a meta-controller for a multi-agent RL control system that dynamically switches between measurement and reset policies to improve scan time while maintaining measurement robustness
- Ideated a multi-objective scheduler that optimizes system throughput while minimizing a secondary metric of interest(i.e. cost, water usage, carbon footprint); bounded tradeoff between metrics using competitive ratios
- Implemented the scheduler on a Kubernetes cluster with Spark, integrating a Python daemon to dynamically scale executor quotas; reduced carbon footprint by up to 33% with minimal increase in makespan

Undergraduate Researcher Aug 2023 – Dec 2023
Applied Materials *Santa Clara, CA*
- Designed a multiple linear regression model with least-squares fitting to approximate wafer topography; outperformed traditional planar and radial models by capturing higher-order spatial variation across the wafer
- Engineered an OpenCV pipeline to detect and segment metal-deposition voids, quantifying size and fill percentage; automated manual inspection process, accelerating quality control to 700ms
- Combined height-profile modeling with image-derived void metrics to deliver lot-level analytics, enabling process engineers to adjust deposition parameters for improved thickness uniformity and yield

Software Engineering Intern May 2023 – Aug 2023
Nokia *Sunnyvale, CA*
- Introduced coverage-analysis software to track instruction-combination execution across single-cycle datapaths; identified gaps in hardware verification for Nokia's next-gen networking chip
- Built visualization tools to evaluate instruction-level coverage at scale, automated generation of new test cases for missing instruction combinations, and deployed scripts into nightly regression runs to quantify testing effectiveness

Selected Publications

- Adam Lechowicz, **Rohan Shenoy**, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Christina Delimitrou, "SPADE: Signal-Aware DAG Scheduling and Dynamic Provisioning for Data Processing Clusters", *In Submission*
- **Rohan Shenoy***, Evan Coleman*, Hans Gaensbauer, Elsa Olivetti, "Counting atoms faster: policy-based nuclear magnetic resonance pulse sequencing for atomic abundance measurement", *Proceedings of ICML 2025*, July 2025.
- Adam Lechowicz, **Rohan Shenoy**, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Christina Delimitrou, "Carbon- and Precedence-Aware Scheduling for Data Processing Clusters", *Proceedings of ACM SIGCOMM 2025*, Short Paper, September 2025
- **Rohan Shenoy**, Hans Gaensbauer, Elsa A. Olivetti, Evan Coleman, "Optimizing NMR Spectroscopy Pulse Sequencing with Reinforcement Learning for Soil Atomic Abundance", *Proceedings of NeurIPS 2024 Workshop on Tackling Climate Change with Machine Learning*, December 2024

Technical Skills

Languages: Python, C/C++, Java, SQL, R, RISC-V, MatLab
Libraries & Frameworks: Pandas, NumPy, gRPC, CVXPy, CV2, Gymnasium, PyTorch, Spark, Kubernetes
Interests: Poker, Skiing, Photography, Backgammon, Soccer